

D. Taiga's Carry Chains

time limit per test: 2 seconds
 memory limit per test: 256 megabytes

*Miracles don't happen to those
 who just wait.*

— Toradora!

After classes at Ohashi High School, Ryuui hands Taiga a non-negative integer n and sets a simple challenge.

They will play for exactly k moves. In a single move, Taiga chooses a non-negative integer ℓ and sets $n \leftarrow n + 2^\ell$.

Ryuui defines the *score* of one move as the number of binary carries that occur when adding 2^ℓ to the current number in base 2. The total *score* is the sum of *score* over all k moves.

Taiga wants the total *score* to be as large as possible after k moves. What is the maximum total *score* she can achieve?

Input

Each test contains multiple test cases. The first line contains the number of test cases t ($1 \leq t \leq 1000$). The description of the test cases follows.

The only line of each test case contains two integers n and k ($1 \leq n < 2^{30}$, $0 \leq k \leq 10^9$) — the initial integer and the number of moves.

Output

For each test case, output a single integer — the maximum total *score* that Taiga can achieve.

Example

input	Copy
6	
7 1	
13 2	
42 2	
1048576 100	
23 2	
371 1	
output	Copy
3	
4	
3	
100	
5	
3	

Note

In the first test case, $(n, k) = (7, 1)$ and $7 = 111_2$. Adding 2^0 gives $111 + 1 = 1000$, which produces carries at bits 0, 1, 2. So the total score is 3.

In the second test case, $(n, k) = (13, 2)$ with $13 = 1101_2$. First we add 2^0 : $1101 + 0001 = 1110$, which creates one carry. Then we add 2^1 : $1110 + 0010 = 10000$,

Codeforces Round 1068 (Div. 2)

Contest is running

01:32:33

Contestant



→ Submit?

Language: GNU G++17 7.3.0

Choose file: No file chosen

Be careful: there is 50 points penalty for submission which fails the pretests or resubmission (except failure on the first test, denial of judgement or similar verdicts). "Passed pretests" submission verdict doesn't guarantee that the solution is absolutely correct and it will pass system tests.

→ Score table

	Score
Problem A	446
Problem B	892
Problem C	1115
Problem D	1561
Problem E	2007
Problem F	2676
Successful hack	100
Unsuccessful hack	-50
Unsuccessful submission	-50
Resubmission	-50

* If you solve problem on 00:27 from the first attempt

with carries propagating through bits 1, 2, 3. In total there are $1 + 3 = 4$ carries, so the score is 4.

In the third test case, $(n, k) = (42, 2)$ and $42 = 101010_2$. First we add 2^1 :

$101010 + 000010 = 101100$, giving one carry. Next we add 2^2 :

$101100 + 000100 = 110000$, which generates carries at bits 2 and 3. Thus the total number of carries is $1 + 2 = 3$.

In the fifth test case, $(n, k) = (23, 2)$ and $23 = 10111_2$. First we add 2^0 :

$10111 + 00001 = 11000$, which produces carries at bits 0, 1, 2. Then we add 2^3 :

$11000 + 01000 = 100000$, producing carries at bits 3 and 4. Altogether there are $3 + 2 = 5$ carries, so the total score is 5.

[Codeforces](#) (c) Copyright 2010-2025 Mike Mirzayanov

The only programming contests Web 2.0 platform

Server time: Dec/05/2025 16:02:25^{UTC+1} (j2).

Desktop version, switch to [mobile version](#).

[Privacy Policy](#) | [Terms and Conditions](#)

Supported by



ITMO